Project Proposal: Sensor-Adaptive Agricultural Vision Foundation Models

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Abstract

This project will develop an agricultural vision foundation model that transfers across crops, field conditions, and sensing modalities. The work will continue self-supervised pretraining of official ImageNet-initialized Vision Transformers (ViT-S, ViT-B, and, where feasible, ViT-L) on a curated agricultural corpus and compare masked image modeling (MIM) with DINOv2-style self-distillation. A learnable 1×1 *spectral adapter will allow the same RGB-pretrained ViT to accept RGB images, GeoTIFF tiles, and multispectral arrays. Success will be assessed through controlled downstream evaluation under realistic domain shift.*

1. Introduction

Agricultural imagery varies widely across crops, phenological stages, farms, seasons, sensors, and acquisition platforms. This variability makes it difficult to train a single model that transfers reliably beyond one benchmark or one data source. At the same time, labels are expensive, and many useful agricultural datasets are fragmented across narrow tasks such as disease classification, weed recognition, seedlings, or crop-type identification. This project asks whether continual self-supervised pretraining can turn official pretrained vision transformers into robust agricultural foundation models that transfer under real deployment conditions.

2. Background and Significance

Generic ViT checkpoints are typically trained on natural RGB imagery and may not encode agricultural morphology, lesion texture, canopy structure, or field context well enough for downstream use. Agricultural sensing also spans close-range RGB photos, UAV imagery, and georeferenced multispectral products such as GeoTIFF. A model that can learn from both RGB and multispectral data would be useful for classification, retrieval, detection, segmentation, and monitoring, while reducing annotation burden across multiple tasks.

3. Data

The verified local corpus currently contains **412,454** image files across **13** downloaded datasets: 400,359 JPG, 11,079 PNG, and 1,010 JPEG files. The current corpus is therefore strong for RGB pretraining, but it still lacks verified local GeoTIFF, near-infrared, and broader multispectral coverage. The next data step is to freeze a provenance-aware RGB release and then extend it with field-scale UAV, Sentinel-2, Landsat, or comparable agricultural sources that include additional spectral bands. For every source, we will record license, citation, version, retrieval date, modality, band metadata, spatial metadata, and exclusion rules before training begins.

4. Project Objectives and Hypotheses

Objective 1: Build a documented agricultural pretraining corpus with source provenance, versioning, and duplicate control.
Objective 2: Pretrain official ViT-S, ViT-B, and ViT-L backbones using MIM and DINOv2-style self-distillation.
Objective 3: Test whether a small spectral adapter enables effective RGB-to-multispectral transfer without replacing the pretrained ViT.
Objective 4: Measure transfer performance under realistic source, crop, geography, season, and sensor shift.

The project will test three hypotheses. First, agricultural continual pretraining will improve low-label and held-out-domain transfer relative to an unadapted ImageNet ViT. Second, reconstruction and self-distillation will provide complementary benefits, making a sequential MIM-to-DINOv2 schedule a competitive candidate. Third, a trainable 1×1 spectral adapter will transfer RGB-pretrained representations to multispectral inputs more effectively than an RGB-only baseline, while adding very few parameters.

5. Research Design and Methodology

Model initialization and sensor adaptation. The project will use official `timm` ViT checkpoints initialized from ImageNet, MAE, DINOv2, or similar pretrained weights rather than random initialization. A learnable 1×1 convolution will project an arbitrary C-band input into the three-channel interface expected by the pretrained ViT. For RGB input, the adapter will be identity initialized. For multispectral input, the available bands will be mapped into the RGB interface while extra band weights start at zero and remain trainable. This preserves the pretrained backbone and gives us a clean ablation against direct patch-embedding expansion.

Continual pretraining objectives. MIM will train the encoder to reconstruct masked patches, encouraging local structure and texture awareness. DINOv2-style training will use a student-teacher self-distillation objective with an exponential-moving-average teacher and multi-crop augmentations to encourage view-invariant representations. We will compare ImageNet-only transfer, MIM-only pretraining, DINOv2-style pretraining, and a sequential MIM-to-DINOv2 schedule.

6. Evaluation Plan and Success Criteria

Experiments will begin with ViT-S on a single RTX 4090 (24 GB) using BF16 mixed precision, gradient accumulation, and gradient checkpointing. Once a stable recipe is identified, we will scale to ViT-B and, where memory permits, ViT-L. Sampling will be source balanced so that large species collections do not dominate disease, weed, or multispectral examples.

Representation quality will be evaluated with linear probing, few-shot adaptation, and full fine-tuning. Classification will report accuracy and macro-F1; compatible detection and segmentation datasets will additionally use mAP and IoU. The primary evaluation protocol will hold out entire sources, crops, farms, geographies, seasons, or sensors. Random image-level splits will be used only as secondary diagnostics. Required baselines are an unadapted ImageNet ViT, RGB-only pretraining, MIM-only training, DINOv2-style training, frozen and trainable adapters, and direct patch-embedding expansion where feasible.

The project will be successful if it shows consistent improvement over generic ViT initialization across multiple transfer settings, stronger performance under limited labels, and credible gains once real multispectral data are incorporated. Supporting evidence will include per-domain metrics, label-efficiency curves, repeated-seed mean and standard deviation, training curves, MIM reconstructions, and student-teacher similarity visualizations.

7. Deliverables, Risks, and Work Plan

The project will produce: a versioned data catalog with provenance and duplicate reports; reproducible MIM and DINOv2-style pretraining configurations for official ViTs; spectral-adapter ablations for RGB and multispectral inputs; domain-shift transfer benchmarks; and training logs, checkpoints, and tracked artifacts suitable for reproducible reruns on another machine.

The principal risks are dataset leakage, RGB-heavy sampling bias, and inadequate multispectral coverage. These risks will be managed through duplicate audits, source-disjoint evaluation, modality-aware manifests, and a clearly scoped conclusion if multispectral acquisition remains incomplete.

Weeks 1–2: freeze corpus version 1, complete provenance and duplicate audits, establish ImageNet ViT baselines, and select a stable ViT-S recipe.
Weeks 3–4: acquire and tile GeoTIFF or multispectral sources, conduct sensor-adapter and objective ablations, and evaluate domain-held-out transfer.
Weeks 5–6: scale the selected recipe, repeat critical experiments, prepare final results, and release the code, manifests, configurations, and permitted checkpoints for submission to a computer-vision or agricultural-informatics venue.

References

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